CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · FORMULA BANK

Chapter 24: Measures of Central Tendency (Mean, Median, Quartiles and Mode)

Every formula and result of the chapter on one sheet

1	Mean of raw data	$\bar{x} = \frac{\sum x}{n}$
2	Mean: direct method	$\bar{x} = \frac{\sum fx}{\sum f}$
3	Mean: short-cut method	$\bar{x} = A + \frac{\sum fd}{\sum f}$, where $d = x - A$
4	Mean: step-deviation method	$\bar{x} = A + \frac{\sum ft}{\sum f} \times i$, where $t = \frac{x - A}{i}$
5	Sum of the observations	$n \times \bar{x}$
6	Median, n odd	The $\left(\frac{n+1}{2}\right)$th term
7	Median, n even	Mean of the $\left(\frac{n}{2}\right)$th and $\left(\frac{n}{2}+1\right)$th terms
8	Lower quartile Q_1	n odd: $\left(\frac{n+1}{4}\right)$th term; n even: $\left(\frac{n}{4}\right)$th term
9	Upper quartile Q_3	n odd: $\left(\frac{3(n+1)}{4}\right)$th term; n even: $\left(\frac{3n}{4}\right)$th term
10	Inter-quartile range	$Q_3 - Q_1$
11	Semi-inter-quartile range	$\frac{Q_3 - Q_1}{2}$
12	Mode	The value with the highest frequency
13	From an ogive	Median at $\frac{N}{2}$, Q_1 at $\frac{N}{4}$, Q_3 at $\frac{3N}{4}$